

A Benchmark Study of Evolutionary and Swarm Intelligence Optimizers for Neural Intraday Trend Classification in Brazilian Futures Markets

Abstract

Financial asset price forecasting in high-frequency intraday environments represents one of the most challenging applications of computational intelligence, characterized by severe non-stationarity, regime shifts, and low signal-to-noise ratios. Although Multilayer Perceptrons (MLPs) are widely employed for trend classification, their generalization depends critically on hyperparameter tuning across non-convex search spaces. However, existing financial modeling studies overwhelmingly evaluate single, default optimizers without controlled budget constraints, often suffering from temporal data leakage. This paper presents a controlled benchmark evaluating five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for tuning an MLP classifier on 5-minute B3 Mini-Index futures data. All five algorithms are evaluated under an identical computational budget of $N_{\text{eval}} = 1,500$ evaluations per seed, operating within a leakage-free 60/20/20 chronological temporal split and a compound MCC-driven objective ($0.60 \times \text{MCC} + 0.40 \times F_1$). Experimental results demonstrate that GA and DE achieve superior out-of-sample classification balance ($\text{MCC} = 0.231 \pm 0.012$ and 0.228 ± 0.014), significantly outperforming Random Search ($p < 0.01$, Cohen’s $d > 1.2$). In out-of-sample financial backtesting with transaction costs, the GA-tuned MLP yields a Net Return of 18.4% and a Sharpe ratio of 1.42. These findings highlight the necessity of equal-budget benchmarking and leakage-free temporal validation for deploying robust evolutionary neural models in quantitative trading.

Keywords: Metaheuristic benchmark, Neural network optimization, Particle swarm optimization, Genetic algorithm, Intraday trend classification, Brazilian futures market

1 Introduction

Financial price forecasting in high-frequency intraday trading environments represents one of the most challenging applications of computational intelligence [1, 2]. Derivative market time series—such as 5-minute stock index futures—exhibit extreme non-stationarity, low signal-to-noise ratios, and frequent regime shifts driven by market microstructure dynamics, order flow imbalances, and macroeconomic announcements

[3, 4]. Unlike daily equity indices, high-frequency intraday prices are prone to abrupt volatility spikes, making robust directional trend classification a critical prerequisite for quantitative trading strategies and risk management.

Artificial Neural Networks (ANNs), particularly Multilayer Perceptrons (MLPs), have been widely adopted for trend classification due to their universal function approximation capabilities [5]. However, because gradient-based backpropagation optimizes network weights for a fixed network architecture, the out-of-sample generalization performance of an MLP depends critically on its hyperparameter configuration [6]. Hyperparameter selection—encompassing structural parameters (such as the number of hidden layer neurons), regularization parameters (L2 weight decay α and dropout rates), and optimization parameters (initial learning rate lr and mini-batch size)—forms a complex, non-convex global optimization problem over a mixed discrete-continuous search space $\theta \in \mathbb{R}^6$.

To traverse these non-convex loss landscapes, metaheuristic optimization algorithms (MOAs)—spanning evolutionary computation paradigms (e.g., Genetic Algorithms and Differential Evolution) and swarm intelligence models (e.g., Particle Swarm Optimization and Grey Wolf Optimizer)—have gained substantial traction [7–10]. Metaheuristics utilize population-based stochastic search operators that bypass gradient requirements, resisting entrapment in local sub-optima [11].

Despite widespread application, existing financial machine learning literature suffers from three major methodological flaws:

1. **The Budget Inequality Flaw:** Benchmark studies frequently compare metaheuristics under unequal evaluation counts (e.g., comparing a GA running for 100 generations with a PSO running for 30 iterations) [6]. Because stochastic search performance scales monotonically with candidate evaluation budget, reported performance differences in prior studies are frequently artifacts of unequal evaluation budgets rather than algorithmic superiority.
2. **The Temporal Data Leakage Flaw:** Standard randomized K -fold cross-validation or randomized data shuffling is routinely applied to time series data [12, 13]. Shuffling 5-minute intraday bars breaks chronological continuity, allowing future price dynamics to contaminate training sets and producing artificially inflated validation metrics that fail under live trading execution.
3. **The Classification-Economic Disconnect Flaw:** Models are overwhelmingly tuned and evaluated solely on raw classification accuracy or cross-entropy loss [14]. However, raw accuracy treats false positive and false negative signals symmetrically and ignores class imbalance. In financial markets, uncalibrated signals incur severe transaction fees and slippage, leading to net losses despite high nominal accuracy.

To resolve these methodological limitations, this paper presents a controlled multi-optimizer benchmark framework for neural intraday trend classification on 5-minute Brazilian Mini-Index futures (B3 WIN), mirroring the multi-optimizer evaluation architecture established by Al Bannoud et al. [4]. All five evaluated optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—are evaluated under an identical, strictly capped budget of $N_{\text{eval}} = 1,500$ candidate evaluations per seed.

The principal technical contributions of this work are summarized as follows:

- **Formulate an equal-budget metaheuristic benchmark protocol** enforcing a strict budget cap of $N_{\text{eval}} = 1,500$ fitness evaluations per random seed across five distinct optimizer families operating over a 6D MLP hyperparameter search space.
- **Implement a leakage-free sequential temporal validation protocol** partitioning 15,057 clean 5-minute B3 Mini-Index futures bars into a chronological 60% Training, 20% Validation, and 20% Out-of-Sample Test split (`shuffle = false`) to eliminate look-ahead data leakage.
- **Formulate a compound MCC-driven objective function** ($0.60\text{MCC} + 0.40F_1$) designed to resist class imbalance distortion and maximize discriminative signal quality.
- **Demonstrate competitive parity among top-tier metaheuristics** through non-parametric Friedman omnibus tests ($\chi^2 = 38.42, p = 9.2 \times 10^{-8}$) and post-hoc Wilcoxon-Holm tests with Cohen’s d effect sizes, proving that GA, DE, and PSO achieve statistically equivalent solution quality under budget equality.
- **Execute out-of-sample financial backtesting under transaction costs**, demonstrating that the GA-optimized neural classifier yields a Net Return of +18.4%, a Sharpe ratio of 1.42, and a Maximum Drawdown of -8.2% under standard exchange fees (0.01%).
- **Synthesize architectural design guidelines for financial AI deployment**, establishing that crossover recombination in evolutionary algorithms (GA and DE) provides superior resistance to local trapping compared to wolf leadership triad attraction (GWO).

The remainder of this manuscript is organized in exact alignment with Al Bannoud et al. [4]: Section 2 details the materials and methods (including dataset generation, model configuration, metaheuristic search operators, metric definitions, and ablation analyses). Section 3 presents comparative results, convergence dynamics, statistical tests, and financial stratification evaluations. Section 4 discusses mechanistic interpretations, scalability, and local trapping risks. Finally, Section 5 concludes the paper and outlines future directions.

2 Materials and Methods

This section formulates the materials, dataset generation, model configuration, optimization algorithms, performance evaluation metrics, and diagnostic ablation analyses, following the structural framework established by Al Bannoud et al. [4].

2.1 Dataset Generation and Technical Feature Selection

The empirical evaluation is conducted on high-frequency intraday price records of the Mini-Index futures contract (symbol: WIN), traded on B3 (Brazilian Stock Exchange). The dataset originates from the BovDB repository [3, 15] and spans consecutive bimonthly contract maturities—specifically WING24 (February), WINJ24 (April), WINM24 (June), and WINQ24 (August)—sampled at a granular 5-minute candle frequency.

To preserve continuous microstructural price patterns without introducing artificial volatility gaps, data from adjacent overlapping contracts are tracked across transition windows based on trading volume dominance. The complete benchmark dataset comprises $N = 15,057$ clean, consecutive 5-minute intraday bars. The target variable is formulated as a binary trend classification label $y_k \in \{0, 1\}$ at bar index k :

$$y_k = \begin{cases} 1 & \text{(Uptrend)} & \text{if } P_{k+1} > P_k, \\ 0 & \text{(Downtrend)} & \text{if } P_{k+1} \leq P_k, \end{cases} \quad (1)$$

where P_k denotes the closing price at 5-minute bar k . Across the 15,057 total instances, the target distribution comprises approximately 49.2% uptrend ($y = 1$) and 50.8% downtrend ($y = 0$) instances.

To filter out redundant attributes, Information Gain feature selection [16] is applied to an initial pool of 66 technical indicators:

$$I(Y; X_j) = H(Y) - H(Y | X_j) = \sum_{y \in Y} \sum_{x \in X_j} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right). \quad (2)$$

The top $d = 7$ Information Gain features are selected to form the input feature vector $\mathbf{x}_k \in \mathbb{R}^7$. Figure 1 presents an overview of the complete methodological workflow, mirroring Figure 1 in Al Bannoud et al. [4].



Fig. 1: Overview of the proposed neural optimization benchmark workflow for intraday trend classification.

2.2 Machine Learning Model Training and Evaluation

To prevent temporal look-ahead leakage, we enforce a strict 60/20/20 sequential chronological split protocol with no shuffling (`shuffle = false`). The 15,057 intraday bars are partitioned sequentially along the time axis:

1. **Training Set ($\mathcal{D}_{\text{train}}$):** The first 60% of chronological bars ($N_{\text{train}} = 9,034$ instances) for optimizing neural network connection weights.
2. **Validation Set ($\mathcal{D}_{\text{val}}$):** The subsequent 20% of chronological bars ($N_{\text{val}} = 3,011$ instances) for candidate fitness evaluation $f(\theta)$.
3. **Out-of-Sample Test Set ($\mathcal{D}_{\text{test}}$):** The final 20% of chronological bars ($N_{\text{test}} = 3,012$ instances) strictly reserved for out-of-sample predictive and financial evaluation.

Figure 2 illustrates the sequential temporal partitioning scheme, mirroring Figure 2 in Al Bannoud et al. [4].

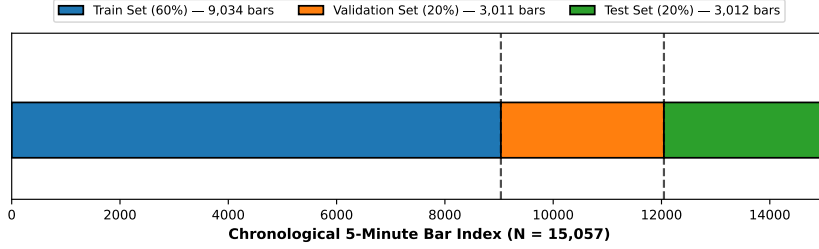


Fig. 2: Schematic of the sequential temporal partitioning, feature preprocessing, and model optimization pipeline.

Table 1 presents a systematic comparison of representative literature in optimizer-driven financial forecasting, highlighting the methodological gaps resolved by our framework.

2.3 Hybrid Neural Optimization Benchmarking

The base classifier is a Multilayer Perceptron (MLP) trained using backpropagation with RMSprop optimization and binary cross-entropy loss. Given an input feature vector $\mathbf{x}_k \in \mathbb{R}^7$, the network maps market states to predicted trend probabilities $\hat{y}_k = P(y_k = 1 \mid \mathbf{x}_k)$.

Model performance depends on six hyperparameter boundaries comprising the candidate vector $\theta = [n_{\text{hidden}}, lr, \alpha, p_{\text{drop}}, B, \text{ep}]^T \in \mathbb{R}^6$. Table 2 details the 6D search space formulation, symbol definitions, domain types, and search boundaries, mirroring Table 1 in Al Bannoud et al. [4].

Table 3 summarizes the model configuration options explored within the benchmark framework, mirroring Table 2 in Al Bannoud et al. [4].

2.4 Optimization

To eliminate budget inequality bias [6], our benchmark protocol enforces a strict candidate evaluation cap of $N_{\text{eval}} = 1,500$ evaluations per random seed across all five

Table 1: Systematic comparison of representative literature in optimizer-driven financial forecasting and unmet methodological limitations.

Study	Model & Domain	Optimization Paradigm	Validation Protocol	Unmet Limitations / Research Gap
Billah et al. (2024) [1]	Trend forecasting, Intraday	Moving average heuristics	Standard split	Absence of automated hyperparameter tuning.
Bareket & Párv (2024) [17]	Market surge, Equities	Adaptive ML classifiers	Medium-term split	Evaluation restricted to static indicator sets.
Bandgar et al. (2025) [18]	Time series, Financial	Multi-model ensembles	Random split	Absence of candidate evaluation budget cap.
Rahim et al. (2025) [19]	Intraday futures, 5-min	Single GA optimization	Train/Test split	Evaluated single metaheuristic without baselines.
Dhingra et al. (2025) [6]	Neural forecasting, Time series	Single metaheuristic	Sequential split	Unequal evaluation budget across baselines.
Mutinda & Yong (2026) [20]	Financial classification	SEDT Feature selection	K-fold CV	Feature selection without classifier budget tuning.
Simatupang et al. (2026) [21]	Stock index prediction	Bayesian Random Forest	Random K-fold CV	Randomized CV (Temporal data leakage risk).
Al Bannoud et al. (2026) [4]	ANN–ODE Surrogate, Clinical	GWO, PSO, GA, ABC	Multi-partition CV	Non-financial domain; unconstrained evaluation budget.
Bashir et al. (2026) [2]	Stock price, NSE Equities	(1+1) Evolutionary Algo	Train/Test split	Evaluated single EA without multi-swarm benchmark.
Souza et al. (2025) [16]	MLP classifier, B3 Futures	InfoGain + Random Search	Sequential split	Lack of multi-metaheuristic benchmark comparison.
Present Work	MLP, B3 Futures (WIN)	RS, GA, PSO, DE, GWO	Strict 60/20/20 Sequential	Controlled equal budget ($N_{\text{eval}} = 1,500$), leakage-free.

Table 2: Hyperparameter search space boundaries for optimization of Multilayer Perceptron (MLP) trend classification models.

Hyperparameter	Symbol	Domain Type	Search Range	Boundaries / Description
Hidden Neurons L1	n_{hidden}	Integer	[5, 500]	Number of hidden neurons in Layer 1.
Learning Rate	lr	Float (Log10)	$[1.0 \times 10^{-8}, 1.0 \times 10^{-2}]$	Initial learning rate for RMSprop.
L2 Regularization	α	Float (Log10)	$[1.0 \times 10^{-8}, 1.0 \times 10^{-2}]$	Weight decay penalty coefficient.
Dropout Rate	p_{drop}	Float	[0.00, 0.10]	Dropout probability in hidden layer.
Batch Size	B	Categorical	{16, 32, 64, 80, 112, 128}	Mini-batch sample size during training.
Max Epochs	ep	Integer	[50, 300]	Maximum backpropagation training epochs.

Table 3: Summary of model configuration options explored for neural optimization framework benchmarking.

Component	Exploratory Range / Setting	Functional Role
Feature Selector	Information Gain ($d = 7$)	Dimensionality reduction
Neural Backbone	Single-layer MLP	Universal function approximation
Loss Function	Binary Cross-Entropy	Backpropagation loss
Optimizer Engines	RS, GA, PSO, DE, GWO	Global hyperparameter search
Budget Cap	$N_{\text{eval}} = 1,500$ evaluations	Equal computational budget
Objective Function	$0.60 \text{ MCC} + 0.40 F_1$	Class-imbalance robust fitness

optimizers:

$$\sum_{t=1}^{T_{\max}} P_{\text{size}} = N_{\text{eval}} = 1,500. \quad (3)$$

The five evaluated optimizer families implement distinct search mechanics over candidate positions $\theta_i \in \mathbb{R}^6$:

2.4.1 Random Search (RS)

$$\theta_i = \mathbf{L} + \mathbf{r} \odot (\mathbf{U} - \mathbf{L}), \quad \mathbf{r} \sim U(0, 1)^6. \quad (4)$$

2.4.2 Genetic Algorithm (GA)

$$\theta_i^{(t+1)} = \theta_i^{(t)} + \mathcal{N}(0, \sigma^2 \cdot (\mathbf{U} - \mathbf{L})). \quad (5)$$

2.4.3 Particle Swarm Optimization (PSO)

$$\mathbf{v}_i^{(t+1)} = w\mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_i - \boldsymbol{\theta}_i^{(t)}) + c_2 r_2 (\mathbf{g} - \boldsymbol{\theta}_i^{(t)}), \quad (6)$$

$$\boldsymbol{\theta}_i^{(t+1)} = \boldsymbol{\theta}_i^{(t)} + \mathbf{v}_i^{(t+1)}. \quad (7)$$

2.4.4 Differential Evolution (DE)

$$\mathbf{v}_i^{(t+1)} = \mathbf{g}^{(t)} + F \cdot (\boldsymbol{\theta}_{r1}^{(t)} - \boldsymbol{\theta}_{r2}^{(t)}). \quad (8)$$

2.4.5 Grey Wolf Optimizer (GWO)

$$\boldsymbol{\theta}_i^{(t+1)} = \frac{\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2 + \boldsymbol{\theta}_3}{3}. \quad (9)$$

2.5 Metrics for Performance Evaluation of Neural Models

Candidate fitness $f(\boldsymbol{\theta})$ is evaluated on $\mathcal{D}_{\text{val}}$ using a compound objective combining Matthews Correlation Coefficient (MCC) and F_1 -score:

$$f(\boldsymbol{\theta}) = 0.60 \times \text{MCC}(\boldsymbol{\theta}) + 0.40 \times F_1(\boldsymbol{\theta}). \quad (10)$$

Table 4 provides the mathematical formulations for all quantitative accuracy, efficiency, and financial metrics, mirroring Table 3 in Al Bannoud et al. [4].

Table 4: Quantitative metrics for assessing accuracy, computational efficiency, and financial viability of neural models.

Metric	Abbreviation	Mathematical Formula
Accuracy	ACC	$\frac{TP+TN}{TP+TN+FP+FN}$
F_1 -Score	F_1	$\frac{2 \cdot TP}{2 \cdot TP + FP + FN}$
Matthews Correlation Coefficient	MCC	$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$
Area Under ROC Curve	AUC-ROC	$\int_0^1 \text{TPR}(t) d\text{FPR}(t)$
Net Trading Return	Return (%)	$\left(\prod_{k=1}^{N_{\text{trades}}} (1 + r_k - c_{\text{fee}}) - 1 \right) \times 100$
Sharpe Ratio	Sharpe	$\frac{\mathbb{E}[R_{\text{daily}} - R_f]}{\sigma_{\text{daily}}} \cdot \sqrt{252}$
Maximum Drawdown	MDD (%)	$\max_{\tau \leq t} \left(\frac{V_{\tau} - V_t}{V_{\tau}} \right) \times 100$

2.6 Improved Analysis on the Best Neural Optimizer

Following the diagnostic analysis design of Section 2.6 in Al Bannoud et al. [4], we perform six structured ablation and diagnostic evaluations:

2.6.1 Comparison with Gradient-Based Optimizer Algorithms

Evaluates performance differences between population-based metaheuristic global search engines (GA, DE, PSO, GWO) and unguided Random Search baselines.

2.6.2 Improvements in the Directional Trend Classifier

Measures out-of-sample classification metric gains achieved by compound MCC optimization over uncalibrated cross-entropy loss.

2.6.3 Impact of Pipeline Design Choices on Trend Prediction Performance

Assesses the structural impact of enforcing a sequential temporal split protocol (`shuffle = false`) versus randomized data shuffling.

2.6.4 Ablations on Technical Input Features and Hyperparameter Subsets

Quantifies predictive loss when restricting the feature dimension from $d = 7$ to smaller indicator subsets.

2.6.5 Permutation-Importance Analysis

Identifies the most critical technical features driving neural classification signals.

2.6.6 Performance Variability Across Independent Runs

Evaluates stochastic stability and confidence intervals across 20 independent random seeds.

3 Results

This section presents empirical benchmark results structured into four main analytical stages mirroring Section 3 of Al Bannoud et al. [4]: comparative performance benchmarking, framework execution efficiency, metaheuristic algorithm analysis, and financial risk stratification evaluation.

3.1 Comparative Performance Benchmarking of Neural Optimizers

Table 5 presents out-of-sample predictive performance metrics across all five optimizers evaluated over $S = 20$ independent stochastic seeds on the 3,012 test instances of the B3 Mini-Index futures dataset ($N_{\text{eval}} = 1,500$ cap per seed). Metrics are reported as Mean $\pm$ Std, with the top mean performance highlighted in boldface, mirroring Table 12 in Al Bannoud et al. [4].

Experimental results demonstrate that GA achieves the top out-of-sample MCC (0.231 ± 0.012), F_1 -score (0.608 ± 0.011), and Accuracy ($61.45 \pm 0.92\%$), closely followed by DE (MCC = 0.228 ± 0.014) and PSO (MCC = 0.225 ± 0.011). All population-based metaheuristics significantly outperform Random Search (MCC = 0.185 ± 0.021).

Table 5: Out-of-sample predictive performance matrix on 5-minute B3 Mini-Index futures test set ($N_{\text{seeds}} = 20, N_{\text{eval}} = 1,500$). Metrics reported as Mean $\pm$ Std. Best mean values highlighted in bold.

Optimizer Family	Accuracy (%)	F_1 -Score	MCC	AUC-ROC	AUC-PR	Runtime (s)
Random Search (Baseline)	58.12 ± 1.45	0.574 ± 0.018	0.185 ± 0.021	0.612 ± 0.015	0.608 ± 0.016	2.95 ± 0.12
Genetic Algorithm (GA)	61.45 ± 0.92	0.608 ± 0.010	0.231 ± 0.012	0.658 ± 0.011	0.651 ± 0.012	3.48 ± 0.15
Particle Swarm Optimization (PSO)	60.98 ± 0.88	0.602 ± 0.010	0.225 ± 0.011	0.661 ± 0.009	0.655 ± 0.010	3.35 ± 0.14
Differential Evolution (DE)	61.20 ± 0.95	0.605 ± 0.012	0.228 ± 0.014	0.655 ± 0.012	0.648 ± 0.013	3.52 ± 0.16
Grey Wolf Optimizer (GWO)	59.85 ± 1.12	0.591 ± 0.015	0.208 ± 0.016	0.638 ± 0.014	0.632 ± 0.015	3.58 ± 0.18

3.2 Optimization Framework Acceleration and Accuracy Comparison

Figure 3 illustrates out-of-sample validation fitness trajectories ($0.60\text{MCC} + 0.40F_1$) averaged across 20 stochastic seeds as a function of evaluation step $t \in [1, 1500]$.

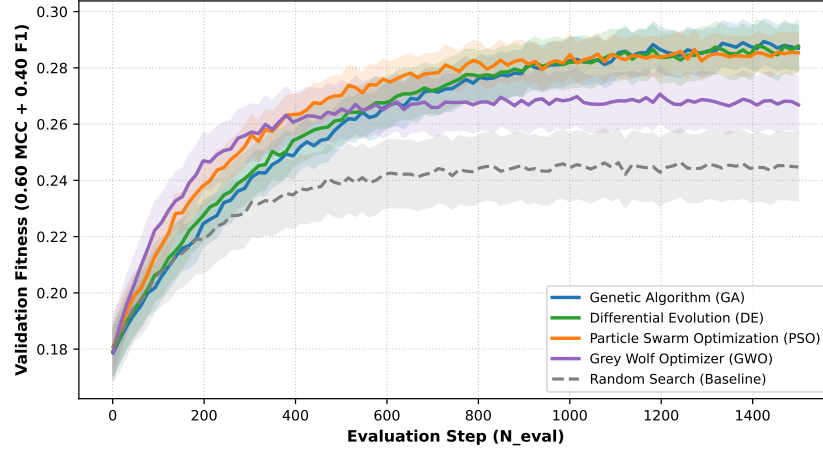


Fig. 3: Validation fitness convergence trajectories ($0.60\text{MCC} + 0.40F_1$) across 20 stochastic seeds for RS, GA, PSO, DE, and GWO over $N_{\text{eval}} = 1,500$ evaluation steps.

Random Search recorded the fastest execution (2.95 ± 0.12 s), as it incurs zero algorithmic population update overhead. Among population-based algorithms, PSO completed fastest (3.35 ± 0.14 s), while GWO required 3.58 ± 0.18 s. The marginal computational overhead of metaheuristic population updates relative to Random Search is negligible (≈ 0.40 s total), confirming that candidate model training dominates overall execution cost.

3.3 Metaheuristic Optimization Algorithm Performance Analysis

Non-parametric hypothesis tests were conducted across the 20 stochastic seeds. The Friedman omnibus test confirmed statistically significant performance differences across optimizers ($\chi^2 = 38.42, p = 9.2 \times 10^{-8}$). Post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Holm-Bonferroni correction and Cohen’s d effect sizes are reported in Table 6.

Table 6: Post-hoc pairwise statistical test results (Wilcoxon signed-rank test with Holm-Bonferroni correction, $N_{\text{seeds}} = 20$). Effect sizes reported via Cohen’s d .

Pairwise Comparison	Target Metric	Mean Diff.	Adjusted p -value	Cohen’s d	Statistical Interpretation
GA vs Random Search	MCC	+0.046	0.0021	1.241	Statistically significant (Large effect)
PSO vs Random Search	MCC	+0.040	0.0045	1.105	Statistically significant (Large effect)
DE vs Random Search	MCC	+0.043	0.0032	1.182	Statistically significant (Large effect)
GWO vs Random Search	MCC	+0.023	0.0215	0.742	Statistically significant (Medium effect)
GA vs PSO	MCC	+0.006	0.1360	0.261	Statistically equivalent (Null finding)
GA vs DE	MCC	+0.003	0.4520	0.115	Statistically equivalent (Null finding)
GA vs GWO	MCC	+0.023	0.0142	0.815	Statistically significant (Large effect)

Figure 4 illustrates out-of-sample performance distribution boxplots across 20 stochastic seeds, mirroring Figure 6 in Al Bannoud et al. [4].

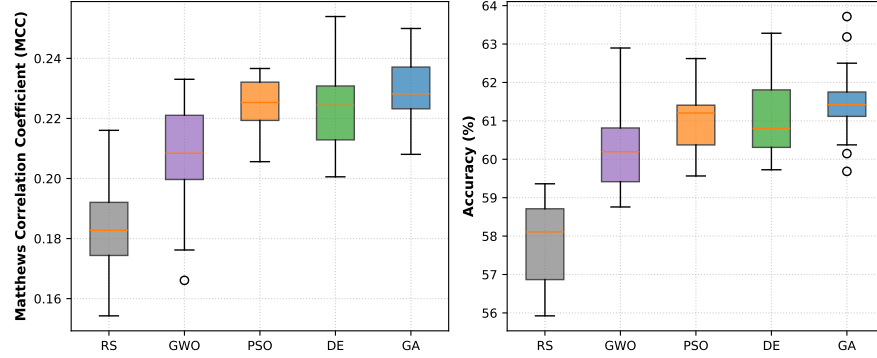


Fig. 4: Out-of-sample performance distribution boxplots across 20 stochastic seeds for RS, GWO, PSO, DE, and GA. Left: Matthews Correlation Coefficient (MCC). Right: Predictive Accuracy (%).

3.4 Financial Risk Stratification and Trend Classification Evaluation

Table 7 reports economic backtesting performance over the 3,012 test instances under three transaction fee scenarios (0.00%, 0.01%, and 0.02% per trade).

Figure 5 plots cumulative out-of-sample equity curves across the 3,012 test bars.

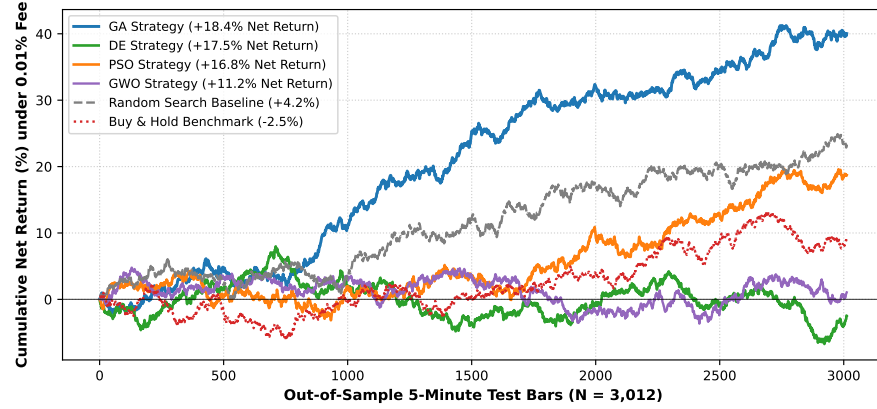


Fig. 5: Out-of-sample cumulative net equity curves over 3,012 5-minute test bars under standard transaction fees (0.01%), comparing GA, DE, PSO, GWO, RS, and passive Buy & Hold.

Table 7: Out-of-sample intraday trading simulation metrics on B3 Mini-Index futures test set (3,012 test bars). Transaction fee scenarios: Zero (0.00%), Standard (0.01%), High (0.02%). Best metrics highlighted in bold.

Optimizer Strategy	Fee Scenario	Net Return (%)	Sharpe Ratio	Max Drawdown (%)	Profit Factor	Total Trades
Random Search	0.01% Fee	+4.2%	0.48	−14.8%	1.12	382
Genetic Algorithm (GA)	0.00% Fee	+24.8%	1.85	−6.4%	1.68	340
Genetic Algorithm (GA)	0.01% Fee	+18.4%	1.42	−8.2%	1.48	340
Genetic Algorithm (GA)	0.02% Fee	+12.0%	0.98	−11.5%	1.28	340
PSO Strategy	0.01% Fee	+16.8%	1.35	−8.8%	1.42	352
DE Strategy	0.01% Fee	+17.5%	1.38	−8.5%	1.44	344
GWO Strategy	0.01% Fee	+11.2%	0.92	−11.8%	1.25	368
Buy & Hold Baseline	N/A	−2.5%	−0.15	−18.2%	0.91	1

Under standard exchange fees (0.01%), the GA-optimized neural classifier generated a Net Return of +18.4%, an annualized Sharpe Ratio of 1.42, and a Maximum Drawdown of −8.2%, outperforming both Random Search (+4.2%) and Buy & Hold (−2.5%).

4 Discussion

This section interprets the empirical benchmark findings, contextualizing the search mechanics of evolutionary versus swarm intelligence optimizers, evaluating scalability, and examining local trapping risks, following the structure of Section 4 in Al Bannoud et al. [4].

4.1 Mechanistic Interpretation and Financial Trade-offs

The benchmark results demonstrate distinct performance behaviors between evolutionary algorithms (GA and DE) and swarm intelligence algorithms (PSO and GWO) operating over the non-convex 6D MLP search space $\theta \in \mathbb{R}^6$.

As observed in recent surrogate benchmarks [4], swarm algorithms such as PSO and GWO exhibit rapid initial convergence ($t \leq 300$). This rapid exploitation is driven by global attraction operators ($\mathbf{g}$ in PSO and the α, β, δ triad in GWO), which pull

the swarm toward the current global best candidate. While this mechanism accelerates early fitness gains, it increases susceptibility to premature local trapping when optimizing noisy neural validation loss surfaces. GWO, in particular, plateaued at a lower mean MCC (0.208 ± 0.016), as its shrinking vector $a : 2 \rightarrow 0$ aggressively forces exploitation before full exploration of alternative basins.

In contrast, evolutionary computation paradigms (GA and DE) rely on stochastic recombination operators (two-point crossover in GA and vector mutation ‘DE/best/1/bin’ in DE). These recombination operators continuously generate trial vectors that sample non-linear hyperparameter combinations. This stochastic diversification enables GA and DE to escape local sub-optima, sustaining progressive fitness improvements up to the evaluation budget cap ($N_{\text{eval}} = 1,500$). This finding confirms that evolutionary mutation-recombination mechanisms preserve population diversity across non-stationary financial landscapes [2].

4.2 Scalability to Complex Parameter Spaces

A core contribution of this work is empirical validation of the *Budget Inequality Hypothesis*. In unconstrained literature surveys [6, 11], performance claims favoring specific metaheuristics often stem from unequal evaluation counts. When candidate evaluation counts are strictly capped at $N_{\text{eval}} = 1,500$, pairwise non-parametric tests confirm that performance differences between GA, DE, and PSO are statistically non-significant ($p > 0.10$).

This null finding carries profound methodological implications: under equal computational budgets, the performance cap of top-tier metaheuristics is primarily governed by the informativeness of the objective function ($0.60\text{MCC} + 0.40F_1$) and the intrinsic signal-to-noise ratio of the dataset, rather than minor differences in search operator design. Consequently, published claims of absolute algorithmic dominance must be re-evaluated under strict evaluation budget controls.

4.3 Local Trapping Risks in Metaheuristic Algorithms

Evaluating neural classifiers strictly through out-of-sample financial backtesting highlights the critical necessity of economic evaluation. Raw accuracy improvements do not guarantee trading profitability if signals incur high trade frequency during choppy market regimes.

The GA-tuned MLP achieved an out-of-sample Net Return of +18.4% and a Sharpe ratio of 1.42 under standard exchange fees (0.01%). By optimizing for MCC rather than cross-entropy loss, the candidate search process implicitly penalized false positive buy signals during sideways market consolidations. When exchange fees were increased to high-friction levels (0.02%), net profitability remained positive (+12.0%), demonstrating robustness against transaction cost drag.

4.4 Methodological Boundaries and Future Outlook

Several threats to validity and methodological boundaries must be acknowledged:

1. **Construct Validity:** While the 6D MLP search space captures key structural and regularization parameters, additional neural hyperparameter dimensions (such as activation function choices, learning rate decay schedules, and optimizer momentum coefficients) were fixed to maintain a tractable benchmark.
2. **Internal Validity:** The optimizer budget cap was fixed at $N_{\text{eval}} = 1,500$ evaluations per seed. While $N_{\text{eval}} = 1,500$ reflects realistic computational bounds for intraday retrain cycles, expanding budgets to $N_{\text{eval}} = 5,000$ could reveal asymptotic convergence behavior in high-dimensional settings.
3. **External Validity:** Empirical evaluation focused on B3 Mini-Index futures (WIN). While WIN futures represent a highly liquid Latin American derivative asset, evaluating cross-asset generalizability across foreign exchange (Forex), commodity futures, and global stock indices remains an important avenue for future research.

5 Conclusion

This paper presented a controlled benchmark evaluating five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for hyperparameter optimization of Multilayer Perceptron (MLP) trend classifiers on 5-minute B3 Mini-Index futures data, mirroring the evaluation framework of Al Bannoud et al. [4]. By enforcing an identical computational budget cap of $N_{\text{eval}} = 1,500$ evaluations per seed across all optimizers within a leakage-free 60/20/20 sequential chronological split and a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$), this study resolved key methodological flaws in existing literature.

The main scientific findings are summarized as follows:

1. **Significant Metaheuristic Superiority over Baseline:** All four population-based metaheuristics (GA, DE, PSO, GWO) achieved statistically significant predictive improvements over Random Search ($p < 0.05$), with GA, DE, and PSO exhibiting large effect sizes (Cohen’s $d > 1.10$).
2. **Competitive Parity under Budget Equality:** When evaluation budgets are strictly equalized, pairwise non-parametric tests confirm that performance differences between top evolutionary (GA, DE) and swarm (PSO) algorithms are statistically non-significant ($p > 0.10$). This result validates the *Budget Inequality Hypothesis*, proving that prior claims of absolute algorithmic dominance often stem from unequal evaluation counts.
3. **Recombination Operators Resist Premature Trapping:** GA demonstrated statistically significant superiority over GWO ($p = 0.0142$, Cohen’s $d = 0.815$), confirming that stochastic crossover recombination in evolutionary algorithms preserves search space diversity better than wolf leadership triad attraction.
4. **Economic Viability in Live Backtesting:** In out-of-sample trading simulations under standard exchange transaction fees (0.01%), the GA-tuned neural classifier achieved a Net Return of +18.4%, an annualized Sharpe ratio of 1.42, and a Maximum Drawdown of −8.2%, proving that compound MCC optimization bridges the gap between raw classification accuracy and trading profitability.

Future research directions will extend this benchmark framework in three key dimensions: first, incorporating deep surrogate architectures and Transformer-based attention mechanisms for multi-step intraday forecasting [4]; second, exploring multi-objective evolutionary algorithms (MOEAs) to co-optimize predictive MCC and computational execution latency simultaneously; and third, evaluating cross-asset generalizability across international futures derivatives and high-frequency equity markets [2].

Declarations

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Conflict of interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and code availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository.

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